National University of Computer and Emerging Sciences

Lahore Campus

Linear Algebra (MT1004)

Date: October 30, 2026

Course Instructor(s)

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Sessional-II Exam Rubric

Total Time (Hrs): 1 Hour

Total Marks: 60

Total Questions: 1

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Roll No **Section** **Student Signature**

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Instruction/Notes:

1. Programmable calculators are not allowed.
2. Marks will be awarded till the correct calculations, incorrect step voids further marking.
3. Attempt all question parts together. Question attempted in separate parts will not be marked.

CLO #2: *Use the properties of vectors in 2-space, 3-space and n-space and recognize vector spaces and/or subspaces to compute their bases and their dimensions.*

Q#1(a)[10]: Find the general solution (X) to the linear system and show that the row vectors of the coefficient matrix are orthogonal to the solution vectors. Explain the geometry of the solution space X? Can we say X is an affine sub-space in R^5?

$$x_1 + 5x_2 + x_3 + 2x_4 - x_5 = 0,$$
$$x_1 - 2x_2 - x_3 + 3x_4 + 2x_5 = 0.$$

Solution:

The augmented matrix of the linear system

$$\begin{bmatrix} 1 & 5 & 1 & 2 & -1 & 0 \\ 1 & -2 & -1 & 3 & 2 & 0 \end{bmatrix}$$ has the reduce row echelon form

$$\begin{bmatrix} 1 & 0 & \frac{-3}{7} & \frac{19}{7} & \frac{8}{7} & 0 \\ 0 & 1 & \frac{2}{7} & \frac{-1}{7} & \frac{-3}{7} & 0 \end{bmatrix}$$

4 Marks

A general solution of the system,

$$x_1 = \frac{3}{7}r - \frac{19}{7}s - \frac{8}{7}t$$

$$x_2 = \frac{-2}{7}r + \frac{1}{7}s + \frac{3}{7}t$$

$$x_3 = r, \quad x_4 = s, \quad x_5 = t$$

Which can be expressed in vector form as

$X = \left(\frac{3}{7}r - \frac{19}{7}s - \frac{8}{7}t, \ \frac{-2}{7}r + \frac{1}{7}s + \frac{3}{7}t, \ r, \ s, \ t\right)$ is orthogonal to the rows of the coefficient matrix of the original system **2 Marks**

And

$$r_1 = (1, 5, 1, 2, -1)$$

$$r_2 = (1, -2, -1, 3, 2) \text{ since}$$

$$r_1 . X = (1)\left(\frac{3}{7}r - \frac{19}{7}s - \frac{8}{7}t\right) + (5)\left(\frac{-2}{7}r + \frac{1}{7}s + \frac{3}{7}t\right) + (1)(r) + (2)(s) + (-1)(t) = 0$$

$$r_2 . X = (1)\left(\frac{3}{7}r - \frac{19}{7}s - \frac{8}{7}t\right) + (-2)\left(\frac{-2}{7}r + \frac{1}{7}s + \frac{3}{7}t\right) + (-1)(r) + (3)(s) + (2)(t) = 0$$ **2 Marks**

The solution space X is a three dimensional affine space R^3 inside R^5. **2 Marks**

(b) [5]: Find the volume of the tetrahedron with vertices $\boldsymbol{P(-1, 2, 0)}$, $\boldsymbol{Q(2, 1, -3)}$, $\boldsymbol{R(1, 1, 1)}$ and $\boldsymbol{S(3, -2, 3)}$.

Solution:

For tetrahedron the volume is

$$\frac{1}{6}\left|\vec{a} . (\vec{b} \times \vec{c})\right|$$

$$\vec{a} = \overrightarrow{PQ}, \ \vec{b} = \overrightarrow{PR}, \ \vec{c} = \overrightarrow{PS}$$

$\overrightarrow{PQ} = (3, -1, -3)$, $\overrightarrow{PR} = (2, -1, 1)$, $\overrightarrow{PS} = (4, -4, 3)$ **1 Marks**

The volume is

$$\frac{1}{6}\left|\overrightarrow{PQ} . (\overrightarrow{PR} \times \overrightarrow{PS})\right|$$

$$= \frac{1}{6}\left| det \begin{bmatrix} 3 & -1 & -3 \\ 2 & -1 & 1 \\ 4 & -4 & 3 \end{bmatrix} \right|$$

3 Marks

$$= \frac{1}{6}|17|$$

$$= \frac{17}{6}$$

1 Mark

c)[10]: Consider the points $\boldsymbol{P(-3,1,0,6)}, \boldsymbol{Q(0,\ 5,1,-2)}$, and $\boldsymbol{R(-4,1,4,0)}$. Find the point $\boldsymbol{S}$ in R^4 whose third component is 6 and such that $\overrightarrow{\boldsymbol{PQ}}$ is parallel to $\overrightarrow{\boldsymbol{RS}}$. Also find the angle between the vectors $\overrightarrow{\boldsymbol{PQ}}$ and $\overrightarrow{\boldsymbol{PR}}$.

Solution:

Denoting $S(a, b, 6, c)$ we have $\overrightarrow{RS} = \overrightarrow{OS} - \overrightarrow{OR} = (a+4, b-1, 2, c)$

For this vector to the parallel to $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = (3, 4, 1, -8)$, there must exist a scalar k such that $\overrightarrow{RS} = k\,\overrightarrow{PQ}$

$(a+4, b-1, 2, c) = k(3, 4, 1, -8)$

3 Marks

By equating both sides we have

$$a+4 = 3k,\ \ b-1 = 4k,\ \ 2 = k,\ \ c = -8k$$

As $k = 2$,So

$$a = 2, \qquad b = 9, \qquad c = -16$$

3 Marks

We conclude that the point S has coordinates $(2, 9, 6, -16)$.

$$\overrightarrow{PQ} = (3, 4, 1, -8)$$

$$\overrightarrow{PR} = (-1, 0, 4, -6)$$

Angle between the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$ is

$$\cos\theta = \frac{\overrightarrow{PQ}.\ \overrightarrow{PR}}{\|\overrightarrow{PQ}\|\|\overrightarrow{PR}\|}$$

2 Marks

$$= \frac{(3)(-1)+(4)(0)+(1)(4)+(-8)(-6)}{\sqrt{3^2+4^2+1^2+(-8)^2}\ \sqrt{(-1)^2+0^2+4^2+(-6)^2}}$$

$$= \frac{49}{\sqrt{90}\sqrt{53}}$$

$$= \frac{49}{3\sqrt{530}}$$

$$\theta = \cos^{-1}\frac{49}{\sqrt{4770}}$$

$$\theta = 44.81$$

2 Marks

d)[10]: Let V be the set of all ordered pairs of real numbers, and consider the following addition and scalar multiplication operations on $\vec{u} = (u_1, u_2)$ and $\vec{v} = (v_1, v_2)$:

$$\vec{u} + \vec{v} = (u_1 + v_1 - 3, u_2 + v_2 + 3), k\vec{u} = (0, ku_2)$$

Find $-\vec{u}$ and $\vec{0}$. Also find vector space axioms that fail to hold.

Solution:

The zero vector $\vec{0}$ must satisfy the following

$$\vec{u} + \vec{0} = \vec{u}$$

Let $\vec{0} = (a, b)$ then

$$(u_1, u_2) + (a, b) = (u_1, u_2)$$

$$(u_1, u_2) + (a, b) = (u_1 + a - 3,\ \ u_2 + b + 3)$$

By comparing both sides we have

$u_1 = u_1 + a - 3$ and $u_2 = u_2 + b + 3$

$a = 3,\ \ b = -3$

So
$\vec{0} = (3, -3)$.

3 Marks

To find the negative vector $-\vec{u}$, it must satisfy the following

$$\vec{u} + (-\vec{u}) = 0$$

$$let\ \vec{u} = (u_1, u_2)$$

$and\ -\vec{u} = (a, b)$ then we have

$$(u_1, u_2) + (a, b) = (3, -3)$$

$$(u_1 + a - 3,\ \ u_2 + b + 3) = (3, -3)$$

By comparing both sides we have

$u_1 + a - 3 = 3$ and $u_2 + b + 3 = -3$

$$a = 6 - u_1$$

$$b = -6 - u_2$$

Hence

$$-\vec{u} = (6 - u_1, -6 - u_2)$$

3 Marks

Axiom (7) Fails:

$$k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$$

L.H.S:

$$k(\vec{u} + \vec{v}) = \big(0,\ \ k(u_2 + v_2 + 3)\big)$$

R. H. S:

$$k\vec{u} + k\vec{v} = (0,\ \ Ku_2) + (0,\ \ kv_2)$$

$$k\vec{u} + k\vec{v} = (-3,\ \ ku_2 + kv_2 + 3)$$

$$L.H.S \neq R.H.S$$

2 Marks

Axiom (8) Fails:

$$(k + m)\vec{u} = \ k\vec{u} + m\vec{u}$$

L.H.S:

$$(k + m)\vec{u} = (0,\ \ (k + m)u_2)$$

R.H.S:

$$k\vec{u} + m\vec{u} = (-3,\ \ ku_2 + mu_2 + 3)$$

1 Mark

$$L.H.S \neq R.H.S$$

Axiom (10) Fails:

$$1\vec{u} = \vec{u}$$

$$1\vec{u} = (0, u_2) \neq \vec{u}$$

1 Mark

where $\vec{u} = (u_1, u_2)$

e)[15]: Consider the bases $B_1 = \{\boldsymbol{u_1}, \boldsymbol{u_2}, \boldsymbol{u_3}\}$ and $B_2 = \{\boldsymbol{v_1}, \boldsymbol{v_2}, \boldsymbol{v_3}\}$ for R^3, where

$$\boldsymbol{u_1} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{u_2} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{u_3} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\boldsymbol{v_1} = \begin{bmatrix} 3 \\ 1 \\ -5 \end{bmatrix}, \quad \boldsymbol{v_2} = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}, \quad \boldsymbol{v_3} = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

i) Find the transition matrices from B_1 to B_2 and B_2 to B_1.

ii) Compute the coordinate vector $[\boldsymbol{w}]_{B_1}$, where $\mathbf{w} = \begin{bmatrix} -5 \\ 8 \\ -5 \end{bmatrix}$, then compute $[\boldsymbol{w}]_{B_2}$.

iii) Check your work by computing $[\boldsymbol{w}]_{B_2}$ directly.

Solution:

i) We have B_1old basis and B_2 is the new basis:

$$P_{B_1 \to B_2} = [B_2 \mid B_1] = \left[\begin{array}{ccc|ccc} 3 & 1 & -1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & -1 & 2 \\ -5 & -3 & 2 & 1 & 1 & 1 \end{array}\right]$$

1 Mark

The reduced row echelon form of this matrix is

$$\left[I \middle| P_{B_1 \to B_2}\right] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3 & 2 & \frac{5}{2} \\ 0 & 1 & 0 & -2 & -3 & \frac{-1}{2} \\ 0 & 0 & 1 & 5 & 1 & 6 \end{array}\right]$$

The transition matrix is $P_{B_1 \to B_2} = \begin{bmatrix} 3 & 2 & \frac{5}{2} \\ -2 & -3 & \frac{-1}{2} \\ 5 & 1 & 6 \end{bmatrix}$

3 Marks

and

$$P_{B_2 \to B_1} = [B_1 \mid B_2] = \left[\begin{array}{ccc|ccc} 2 & 2 & 1 & 3 & 1 & -1 \\ 1 & -1 & 2 & 1 & 1 & 0 \\ 1 & 1 & 1 & -5 & -3 & 2 \end{array}\right]$$

The reduced row echelon form of this matrix is

$$\left[I \middle| P_{B_2 \to B_1}\right] = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{35}{2} & \frac{19}{2} & \frac{-13}{2} \\ 0 & 1 & 0 & \frac{-19}{2} & \frac{-11}{2} & \frac{7}{2} \\ 0 & 0 & 1 & -13 & -7 & 5 \end{array}\right]$$

The transition matrix is $P_{B_2 \to B_1} = \begin{bmatrix} \frac{35}{2} & \frac{19}{2} & \frac{-13}{2} \\ \frac{-19}{2} & \frac{-11}{2} & \frac{7}{2} \\ -13 & -7 & 5 \end{bmatrix}$

3 Marks

ii) Expressing w as a linear combination of u_1, u_2 *and* u_3 we obtain

$$\begin{bmatrix} -5 \\ 8 \\ -5 \end{bmatrix} = c_1 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Equating corresponding components on both sides yields the linear system

$2c_1 + 2c_2 + c_3 = -5$

$c_1 - c_2 + 2c_3 = 8$

$c_1 + c_2 + c_3 = -5$

whose augmented matrix has the reduced row echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 9 \\ 0 & 1 & 0 & -9 \\ 0 & 0 & 1 & -5 \end{bmatrix}$$

The solution of the linear system is

$c_1 = 9,\ \ c_2 = -9,\ \ c_3 = -5$

3 Marks

therefore the coordinate vector is $[\boldsymbol{w}]_{B_1} = \begin{bmatrix} 9 \\ -9 \\ -5 \end{bmatrix}$

Using $[\boldsymbol{w}]_{B_2} = P_{B_1 \to B_2}[\boldsymbol{w}]_{B_1}$

$$= \begin{bmatrix} 3 & 2 & \frac{5}{2} \\ -2 & -3 & \frac{-1}{2} \\ 5 & 1 & 6 \end{bmatrix} \begin{bmatrix} 9 \\ -9 \\ -5 \end{bmatrix} = \begin{bmatrix} \frac{-7}{2} \\ \frac{23}{2} \\ 6 \end{bmatrix}$$

2 Marks

iii) Expressing w as a linear combination of $u_1',\ u_2',\ u_3'$ we obtain

$$\begin{bmatrix} -5 \\ 8 \\ -5 \end{bmatrix} = c_1 \begin{bmatrix} 3 \\ 1 \\ -5 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}$$

Equating corresponding components on both sides yields the linear system

$3c_1 + c_2 - c_3 = -5$

$c_1 + c_2 \qquad = 8$

$-5c_1 - 3c_2 + 2c_3 = -5$

whose augmented matrix has the reduced row echelon form

$$\begin{bmatrix} 1 & 0 & 0 & \frac{-7}{2} \\ 0 & 1 & 0 & \frac{23}{2} \\ 0 & 0 & 1 & 6 \end{bmatrix}$$

The solution of the linear system is

$$c_1 = \frac{-7}{2},\ \ c_2 = \frac{23}{2},\ \ c_3 = 6$$

therefore the coordinate vector is $[\boldsymbol{w}]_{B_2} = \begin{bmatrix} \frac{-7}{2} \\ \frac{23}{2} \\ 6 \end{bmatrix}$

3 Marks

f)[10]: Find the coordinate vector of $\boldsymbol{p}$ relative to the basis $S = \{\boldsymbol{p_1}, \boldsymbol{p_2}, \boldsymbol{p_3}\}$ for $\boldsymbol{P_2}$.

$$\boldsymbol{p} = 2 - x + x^2; \boldsymbol{p_1} = 1 + x, \boldsymbol{p_2} = 1 + x^2, \boldsymbol{p_3} = x + x^2.$$

Solution:

Expressing P as a linear combination of p_1, p_2 and p_3 we obtain

$$2 - x + x^2 = c_1(1 + x) + c_2(1 + x^2)x + c_3(x + x^2)$$

2 Marks

Grouping the terms on the right hand side according to powers of x yields

$$2 - x + x^2 = (c_1 + c_2) + (c_1 + c_3)x + (c_2 + c_3)x^2$$

For this equality to hold for all x, the coefficients associated with the same power of x on both sides must match. This leads to the linear system

$$\begin{aligned} c_1 + c_2 \quad &= 2 \\ c_1 \quad + c_3 &= -1 \\ c_2 + c_3 &= 1 \end{aligned}$$

2 Marks

whose augmented matrix has the reduced row echelon form

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -1 \end{bmatrix}$$

The solution of the linear system is

$$c_1 = 0, \ c_2 = 2, \ c_3 = -1$$

4 Marks

therefore the coordinate vector is $(p)_s = (0, \ 2, -1)$.

2 Marks